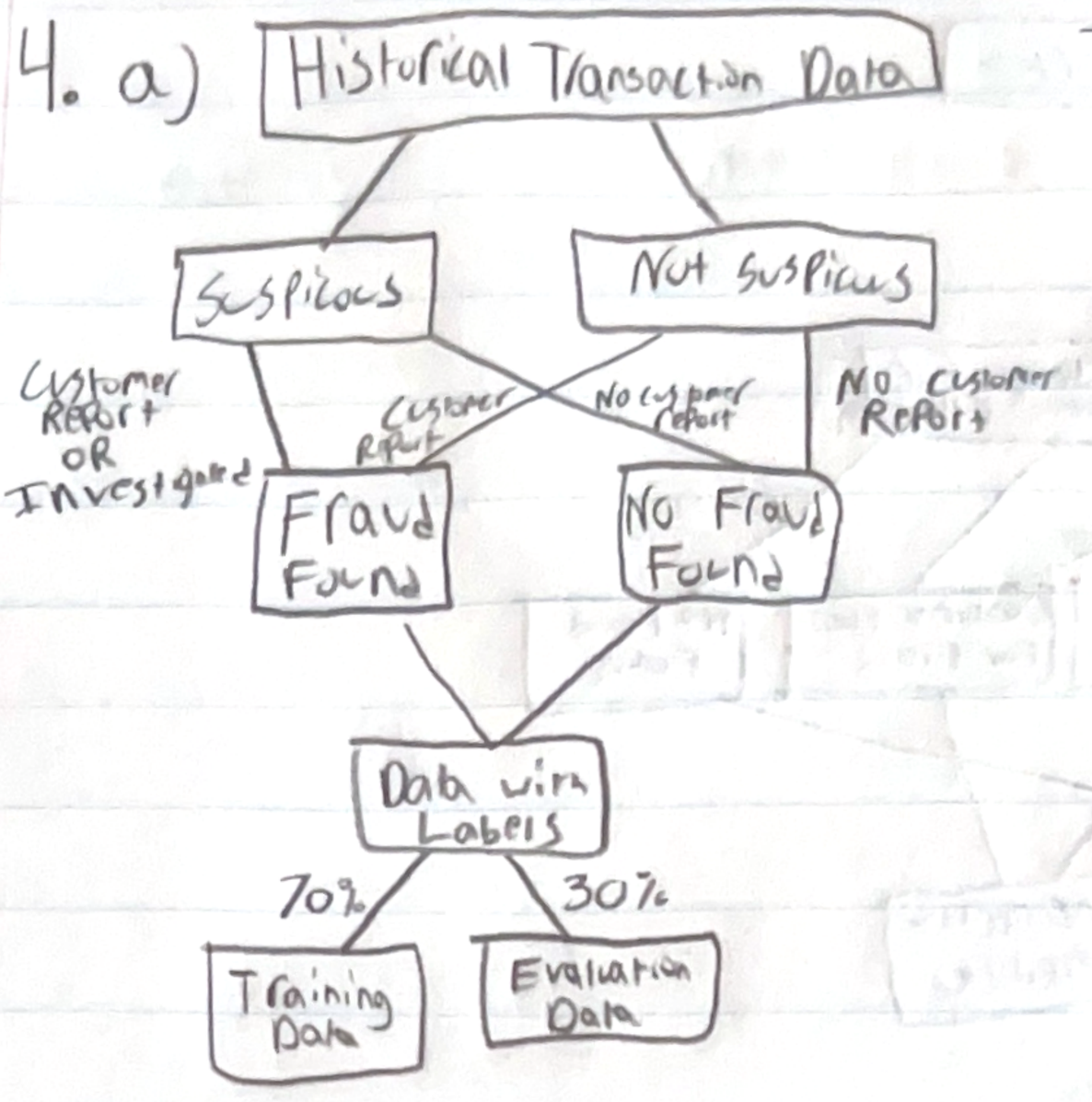
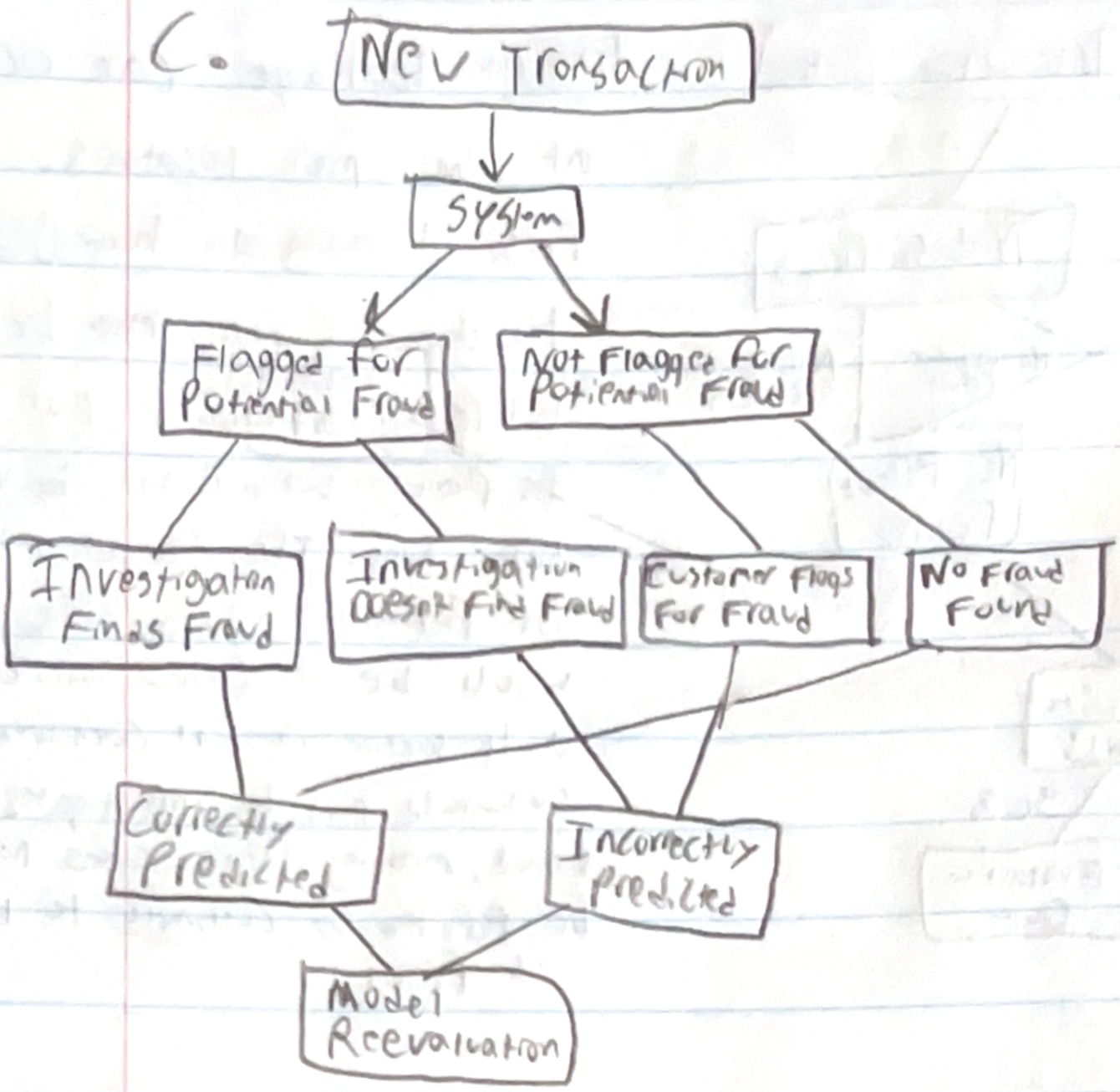
4. a)



Target leakage can occur at two main locations. First depending on how old the data is, there may be cases where a transaction is not labelled as fraud, when it will be at a later time. The second example of potential data leakage would be in cases where a transaction is not considered suspicous, so would not be investigated for fraud, meaning these cases need to be reported by customers to be flagged as fraud.

b) Candidate models are developed by working with the data to develop new features, run the data on multiple different models and tune hyperparameters to develop the models. To evaluate the models, test the models on the evaluation data, then using various performance metrics to evaluate the models. Finally, to compare the models the performance metrics and types of errors can be compared to determine the model best fit for the problem. Before a model is applied for deployment, it must meet certain performance metrics and be deemed the correct model for the situation, both analytically and computationally.

c.



d. At the time of the prediction purchase information including location, item, time, and price would be available. In addition other past transactions by the card holder would be available. What would not be available when the prediction is made is investigation information and customer reporting because these are not determined till later.

e. Customer reports and investigator decisions will catch new cases of fraud that can help to train the model in the future. In cases that are not investigated, it is very unlikely that fraud will be detected. This will cause missed cases of fraud that will result in these cases being incorrectly labeled in future data collection, skewing future models.